LIFTS OF AN OBJECT FORM AN FPPF SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1606-1618 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 1620-1696, the complete author proof (75 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 13 of queue_stacks-stacks-more-morphisms.json (pb=3185, stmt 1606-1618, proof 1620-1696).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned. Named omission: the 2-cocycle identity check is standard bookkeeping on the displayed mechanism (artin #13 precedent).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 07YB: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1696–1696. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03FJ: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1696–1696. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-inf-quasi-coherent: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1566–1572. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 021S: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1696–1696. Cited by the retained statement or

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack over a base scheme S. Assume $\mathcal{I}_\mathcal{X} \to \mathcal{X}$ is locally of finite presentation. Let $(A' \to A, x)$ be a deformation situation. Then the functor*

$$F : B' \longmapsto \{\textit{lifts of } x|_{B' \otimes_{A'} A} \textit{ to } B'\}/\textit{isomorphisms}$$

is a sheaf on the site $(\mathit{Aff}/\operatorname{Spec}(A'))_{fppf}$ of Topologies, Definition 021S.

Proof. Let $\{T'_i \to T'\}_{i=1,\ldots n}$ be a standard fppf covering of affine schemes over A'. Write $T' = \operatorname{Spec}(B')$. As usual denote

$$T'_{i_0 \ldots i_p} = T'_{i_0} \times_{T'} \ldots \times_{T'} T'_{i_p} = \operatorname{Spec}(B'_{i_0 \ldots i_p})$$

where the ring is a suitable tensor product. Set $B = B' \otimes_{A'} A$ and $B_{i_0 \ldots i_p} = B'_{i_0 \ldots i_p} \otimes_{A'} A$. Denote $y = x|_B$ and $y_{i_0 \ldots i_p} = x|_{B_{i_0 \ldots i_p}}$. Let $\gamma_i \in F(B'_i)$ such that γ_{i_0} and γ_{i_1} map to the same element of $F(B'_{i_0 i_1})$. We have to find a unique $\gamma \in F(B')$ mapping to γ_i in $F(B'_i)$.

Choose an actual object y'_i of $\mathit{Lift}(y_i, B'_i)$ in the isomorphism class γ_i. Choose isomorphisms $\varphi_{i_0 i_1} : y'_{i_0}|_{B'_{i_0 i_1}} \to y'_{i_1}|_{B'_{i_0 i_1}}$ in the category $\mathit{Lift}(y_{i_0 i_1}, B'_{i_0 i_1})$. If the maps $\varphi_{i_0 i_1}$ satisfy the cocycle condition, then we obtain our object γ because $\mathcal{X}$ is a stack in the fppf topology. The cocycle condition is that the composition

$$y'_{i_0}|_{B'_{i_0 i_1 i_2}} \xrightarrow{\varphi_{i_0 i_1}|_{B'_{i_0 i_1 i_2}}} y'_{i_1}|_{B'_{i_0 i_1 i_2}} \xrightarrow{\varphi_{i_1 i_2}|_{B'_{i_0 i_1 i_2}}} y'_{i_2}|_{B'_{i_0 i_1 i_2}} \xrightarrow{\varphi_{i_2 i_0}|_{B'_{i_0 i_1 i_2}}} y'_{i_0}|_{B'_{i_0 i_1 i_2}}$$

is the identity. If not, then these maps give elements

$$\delta_{i_0 i_1 i_2} \in \operatorname{Inf}_{y_{i_0 i_1 i_2}}(J_{i_0 i_1 i_2}) = \operatorname{Inf}_y(J) \otimes_B B_{i_0 i_1 i_2}$$

Here $J = \operatorname{Ker}(B' \to B)$ and $J_{i_0 \ldots i_p} = \operatorname{Ker}(B'_{i_0 \ldots i_p} \to B_{i_0 \ldots i_p})$. The equality in the displayed equation holds by Lemma lemma-inf-quasi-coherent applied to $B' \to B'_{i_0 \ldots i_p}$ and y and $y_{i_0 \ldots i_p}$, the flatness of the maps $B' \to B'_{i_0 \ldots i_p}$ which also guarantees that $J_{i_0 \ldots i_p} = J \otimes_{B'} B'_{i_0 \ldots i_p}$. A computation (omitted) shows that $\delta_{i_0 i_1 i_2}$ gives a 2-cocycle in the Čech complex

$$\prod \operatorname{Inf}_y(J) \otimes_B B_{i_0} \to \prod \operatorname{Inf}_y(J) \otimes_B B_{i_0 i_1} \to \prod \operatorname{Inf}_y(J) \otimes_B B_{i_0 i_1 i_2} \to \ldots$$

By Descent, Lemma 03FJ this complex is acyclic in positive degrees and has $H^0 = \operatorname{Inf}_y(J)$. Since $\operatorname{Inf}_{y_{i_0 i_1}}(J_{i_0 i_1})$ acts on morphisms (Artin's Axioms, Remark 07YB) this means we can modify our choice of $\varphi_{i_0 i_1}$ to get to the case where $\delta_{i_0 i_1 i_2} = 0$.

Uniqueness. We still have to show there is at most one γ restricting to γ_i for all i. Suppose we have objects y', z' of $\mathit{Lift}(y, B')$ and isomorphisms $\psi_i : y'|_{B'_i} \to z'|_{B'_i}$ in $\mathit{Lift}(y_i, B'_i)$. Then we can consider

$$\psi_{i_1}^{-1} \circ \psi_{i_0} \in \operatorname{Inf}_{y_{i_0 i_1}}(J_{i_0 i_1}) = \operatorname{Inf}_y(J) \otimes_B B_{i_0 i_1}$$

Arguing as before, the obstruction to finding an isomorphism between y' and z' over B' is an element in the H^1 of the Čech complex displayed above which is zero. □